Copyright © 2023 by the author(s). Published here under license by the Resilience Alliance.
Falk, T., W. Zhang, R. Meinzen-Dick, L. Bartles, R. Sanil, P. Priyadarshini, and I. Soliev. 2023. Games for experiential learning: triggering collective changes in commons management. Ecology and Society 28(1):30. https://doi.org/10.5751/ES-13862-280130

Research

Games for experiential learning: triggering collective changes in commons management

Thomas Falk [1,2], *Wei Zhang* [1], *Ruth Meinzen-Dick* [1], *Lara Bartels* [3,4], *Richu Sanil* [5], *Pratiti Priyadarshini* [5] *and Ilkhom Soliev* [6]

ABSTRACT. As resource users interact and impose externalities onto each other, institutions are needed to coordinate resource use, create trust, and provide incentives for sustainable management. Coordinated collective action can play a key role in enabling communities to manage natural resource commons more sustainably. But when such collective action is not present, what can be done to foster it? We contribute to the understanding of how experiential learning through games can affect behavioral change, potentially leading to more sustainable commons management. We present a conceptual framework describing the most important processes involved in experiential learning games. The framework highlights the importance of the game context for achieving game outcomes. We list game features that have been argued to influence learning and behavioral determinants, focusing on the game narrative and experience, game rules, and attributes of players. We briefly describe how each game feature influences the processes in the framework. Next, we apply the conceptual framework to examine design features that were particularly important for influencing behavioral drivers in commons management in three intervention cases from India relating to groundwater, surface water, and forests. Our conceptual reflections underpin the need to debate about underlying assumptions in using games as intervention tools. Making assumptions transparent can help to understand why or under what conditions experiential learning works or fails. There is a critical need for more systematic choices of the right tools for the right purpose. This includes participants of the experiential learning who must be able to relate the game to their real life. A social dilemma in the game should at least, in its basic structure, represent a real-life dilemma. We close by highlighting future research needs both from the conceptual behavioral change as well as the game design perspective.

Key Words: *behavioral change; facilitation tools; forest; India; social learning; water*

INTRODUCTION

Although key factors for successful commons governance lie in resource users' intrinsic motivation and self-sustained collective governing mechanisms (Ostrom 1990), it is difficult for external actors to support these factors. We explore the potential of one approach to catalyze community-driven processes using experiential learning through games to trigger collective behavioral and institutional change. Emerging evidence on the potential of games to support behavioral change in the management of shared resources, herein broadly referred to as commons management, merits greater attention and investment in research and pilot implementation.

To be effective, rules for commons management should be adapted to the specific context (Ostrom 2005, 2007). This makes it challenging for policy makers and other actors to intervene at scale, however. Billions of dollars have been spent on projects to improve commons management, but sustainable resource management often does not emerge or breaks down after projects end. Although examples of successful facilitation interventions exist to support communities to craft locally adapted rules, these are often implemented with limited reach (Kolavalli and Kerr 2002). Participatory approaches hold promise to ensure solutions, and many solution-seeking processes are thus community driven (Pahl-Wostl and Hare 2004). The key question, therefore, is how to support community-driven coordination, rules, and behavior without external imposition and in a low-cost manner that supports large-scale uptake.

An emerging body of literature explores the use of group dynamic games as an intervention tool to facilitate sustainable commons management (Becu et al. 2017, Craven et al. 2017, Ferrero et al. 2018, Meinzen-Dick et al. 2018, Meyer et al. 2021, Bartels et al. 2022) because they provide participants with useful metaphors for their daily lives (Cardenas and Carpenter 2008, Meinzen-Dick et al. 2018). Approaches cover role-playing games (Barreteau et al. 2001, Bousquet et al. 2002) as well as behavioral games used in experimental economics (Cardenas and Carpenter 2008, Janssen et al. 2011a). We mainly refer to games that are played face-to-face with community members and leaders. There is growing recognition that games can offer a space for experiential learning, which includes experiencing, reflecting, and experimenting (Kolb and Kolb 2009). The approach contrasts with more formal presentations of abstract knowledge that are common in many forms of teaching or costly learning-by-doing in real life (Hertzog et al. 2014, Flood et al. 2018). Games create a relatively low-risk forum for experiencing and discussing the complexities of social-ecological systems and allow for exploring behavior that is too risky in real life (Starks 2014, Speelman et al. 2018, Muhamad and Kim 2020). Using games can lead directly to inventing and negotiating rules, including enforcement mechanisms (Woodhill 2010). Players can experiment with rule making (Speelman et al. 2018) and can take this experience to interact with the wider community. Because the exercise can trigger constructive interaction of resource users and other stakeholders, the resulting social learning effect can potentially go beyond the specific issue or framing of the game to support the development of institutional capacity for sustainable commons management (Meinzen-Dick et al. 2018).

As games gain increasing attention as an intervention tool, there is a danger of designing game-based interventions based on simple and unclear assumptions about human behavior. Den Haan and van der Voort's (2018) systematic review of 42

[1]International Food Policy Research Institute (IFPRI), [2]Philipps-University of Marburg/Germany, [3]Leibniz Centre for European Economic Research (ZEW)/Germany, [4]University of Kassel/Germany, [5]Foundation for Ecological Security/India, [6]Martin Luther University of Halle-Wittenberg/Germany

publications on serious games and Flood et al.'s (2018) systematic review of 43 publications on serious games, in the climate change context, found that games are often assumed to influence behavior through three kinds of learning: cognitive (new or restructured knowledge), normative (change in norms, values, and group opinion), and relational (better understanding of others, building relationships, and trust). However, most interventions and evaluations of collaborative games focus on cognitive learning. Less attention is paid to normative and relational learning (Baird et al. 2014), which are important for triggering behavioral change in sustainable commons management.

We respond to this observation by focusing on experiential learning games, which explicitly address social dilemmas around commons management. Based on previous work, we (1) conceptualize how games can influence behavior related to commons management, (2) identify design elements that support this process best, and (3) reflect on game intervention cases. With that, we contribute to the understanding of how experiential learning through games can affect behavioral change, potentially leading to more sustainable commons management. We present a conceptual framework describing the most important processes involved in experiential learning games. The framework highlights the importance of the game context for achieving game outcomes. We list game features that have been argued to influence learning and behavioral determinants, focusing on the game narrative and experience, game rules, and attributes of players. We briefly describe how each game feature influences the processes in the framework. Next, we apply the conceptual framework to examine design features that were particularly important for influencing behavioral drivers in commons management in three intervention cases from India. Although these cases do not cover all aspects of the framework, we initiate a more systematic discussion on the use of experiential games for behavioral change in the context of commons management. Our reflections allow us to draw conclusions about methodological and operational implications, which can help make games more effective and efficient in triggering behavioral change toward sustainable commons management. In this way we respond to the call to provide better guidance for the design of experiential learning interventions (Flood et al. 2018, Speelman et al. 2018).

METHODS

Three situated case studies of games in the context of facilitating sustainable commons management in India, were instrumental in developing our conceptual thinking and more strongly integrating this thinking into game design. Between 2013 and 2018, the three cases were designed, tested, and implemented by the same partners: the Foundation for Ecological Security (FES), the International Food Policy Research Institute (IFPRI), and the International Crops Research Institute for the Semi-Arid Tropics (ICRISAT). All three game cases were published (Meinzen-Dick et al. 2016, 2018, Falk et al. 2019, Zhang et al. 2021, Bartels et al. 2022). Previous publications focused, however, on the dynamics within the games or the impacts the games achieved. The partners were also driven by the question of how game features influence individual and collective factors expected to change real-life behavior. Thus, we focus on the game design, supplementing quantitative impact assessments with feedback from communities and field staff, and discuss it in light of our conceptual thinking on experiential learning.

For conceptualizing how games can influence behavior, we combine qualitative insights from our practical game experiences with feedback from communities, field staff, and findings from other related studies. In all phases of the game design, testing and revising experiences were collectively reflected on and incorporated into the conceptual framework and the design of the games. Lessons learned with one game contributed to refinements in another of the three games. Feedback during international conferences was part of this process. We complemented this practical reflection with a continuous conceptual review of related theories and empirical evidence, resulting in the version of the conceptual framework presented below.

Our analysis demonstrates an approach of critical reflection, which can contribute to conceptualization based on situated knowledge. Being transparent about the unavoidable subjective perspective analysis of our cases, we contribute to designing more powerful intervention tools through informed arguments (Greenhalgh et al. 2018). Our experience-based insights demonstrate that the conceptual framework can guide and structure reflections on intervention designs from a behavioral change perspective.

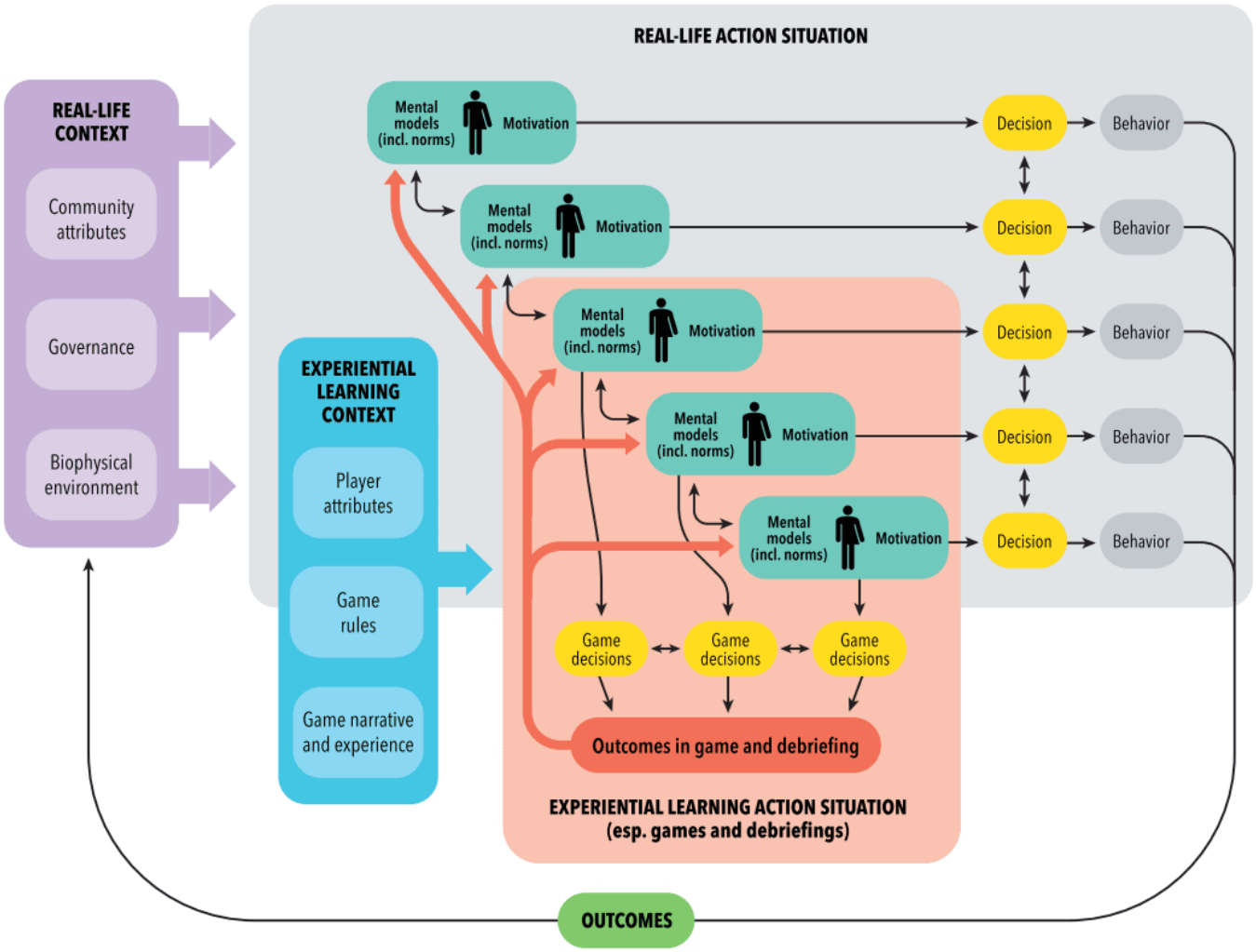
CONCEPTUALIZING HOW GAMES CAN INFLUENCE BEHAVIOR

To address our first objective, we present a framework representing a theory of change, expressing assumptions about causal relationships, and formulating expectations about the mechanisms through which experiential learning leads to real-life behavioral changes (Flood et al. 2018, den Haan and van der Voort 2018, Muhamad and Kim 2020; Fig. 1). The framework is inspired by the institutional analysis and development (IAD) framework (Ostrom 2011), which helps to understand complex collective action problems as well as the deliberative value formation framework (Kenter et al. 2016a), which conceptualizes how deliberation leads to the formation of shared values. We integrate lessons from psychology, economics, and behavioral and education science related to individual and collective learning and behavior.

Our analytical focus is on action situations, which are structured social spaces where actors learn, make decisions, and interact to achieve outcomes (Ostrom 2011). Our framework distinguishes between the real-life action situation (RLAS; gray box in Fig. 1) and the experiential learning action situation (ELAS, apricot box in Fig. 1). The RLAS can be any social-ecological interaction and is embedded in a broader real-life context (purple box in Fig. 1). Community attributes, governance arrangements, and the biophysical environment are important real-life context dimensions (Ostrom 2011, Muhamad and Kim 2020).

In our case, the ELAS is built around games but also includes debriefings in which game players and other community members share their experiences and discuss connections to their real life. An overlap exists between the actors present in the ELAS and the RLAS. Our underlying assumption is that experiences in the ELAS influence behavioral determinants, which then affect actors' behavior in the RLAS (Ostrom 2009, Parks et al. 2013, den Haan and van der Voort 2018). More specifically, we assume that outcomes experienced in the ELAS create feedback to mental models and motivation as two behavioral determinants, which are

Fig. 1. Conceptualizing change in human behavior through experiential learning.



Experiential learning context: Interplay of design features that create and structure the game experience.

Player attributes: Composition of game players, especially in terms of identities, livelihoods, motivations, and mental models.

Game rules: Diversity of rules defining and constraining the decisions to be taken and outcomes experienced in the game.

Game narrative and experience: Narrative in which game rules are embedded and game features that influence the experience of playing.

Mental models: Internal cognitive structures about underlying (a) biophysical and (b) social dynamics that include descriptive and injunctive norms.

Motivations: Diversity of human needs stimulating actions; needs for security, self-determination, competence, belonging, entertainment, and subsistence are important in our context.

Community attributes: Group composition, especially in terms of identities, livelihoods, motivations, and mental models of actors involved in real-world management challenge.

Governance: Interactions among formal and informal rules and processes that shape how people make decisions, share power, and enforce agreed rules.

Biophysical environment: State of the (a) shared resource and (b) infrastructure.

Decision making: Both (a) habitual and (b) deliberative decision making.

Behavior: Includes (a) appropriation, (b) resource provisioning, and (c) institutional service provisioning action.

Experiential learning: A process of co-designing innovations that involves concrete experience, reflective observation, abstract conceptualization, and active experimentation.

Outcomes in game and debriefing: In-game consequences of game decisions defined by game rules that create feedback, especially to mental models.

Action situations: A structured social space where actors interact in achieving outcomes.

Outcomes: Real-world consequences of actors' behavior that create feedback to the context.

also relevant in the RLAS. The quality of this feedback depends on the experiential learning context (blue box in Fig. 1). Important domains comprise player attributes, game rules, game narrative, and experience.

A first channel through which experiential learning can influence real-life behavior is by changing mental models and norms in a participatory way at the individual level (green boxes in Fig. 1). This can lead to stronger convergence of mental models and norms among participants (Wouters et al. 2013, Scholz et al. 2014, Starks 2014, den Haan and van der Voort 2018, Flood et al. 2018, Muhamad and Kim 2020, Garcia et al. 2022). Such a process can be called social learning (Scholz et al. 2014).

Mental models refer to functional but incomplete internal cognitive representations of systems, including beliefs about ecological and technical components of the system as well as perceptions about the constellation of stakeholders, their perspectives, resources, motives, and roles in the governance framework (Jones et al. 2011, Shelton et al. 2018, Garcia et al. 2022). In many cases, actors struggle to sense the impact of their actions on themselves and others at different spatial and temporal scales (van Vugt et al. 2014). Games can help to condense such interactions and create awareness for system dynamics and causal links between own action and outcomes (Flood et al. 2018, Speelman et al. 2018).

Norms are an important aspect of mental models. Behavior is influenced by both descriptive norms (when individuals witness what others do) and injunctive norms (when individuals form perceptions of whether their actions will be approved by others [Cialdini 2003]). Often people take actions based on assumptions about perceived norms in a specific community or broader society. The social interaction in the ELAS can help to confirm or question such assumptions, which supports the formation and expression of shared values and norms (Kenter et al. 2016a, den Haan and van der Voort 2018). Experiential learning can activate transcendental values and trigger a change in more specific norms (Kenter et al. 2016a). Participants first discuss what should be done in the ELAS and then take this discussion to the wider community in reference to the RLAS. Players in the ELAS experience how others act on such norms and how they accept jointly formulated rules. This can support changing expectations about the real-life behavior of others as parts of mental models and the foundation for social norms (Balliet 2010, Hertzog et al. 2014, Meinzen-Dick et al. 2016), which can contribute to trust building (den Haan and van der Voort 2018, Flood et al. 2018).

Motivation is another important behavioral determinant. It describes the diversity of human needs and the desire to fulfill them, which stimulates actions. Different need classifications have been proposed (Maslow 1981, Max-Neef 1991). In the context of experiential learning, the needs for security, self-determination, competence, belonging, entertainment, and subsistence are most relevant (Muhamad and Kim 2020, DeCaro et al. 2021). For instance, self-determination and belongingness needs are activated if people can successfully solve collective action challenges in the ELAS (Muhamad and Kim 2020). Experiencing self-efficacy in the ELAS can influence players' belief in their personal competence to solve challenges in the RLAS (Starks 2014). The less obtrusive, self-controlled, and pro-social approach in the ELAS can stimulate different motivational factors and thus reduce resistance to change often observed in more conventional teaching practices (Muhamad and Kim 2020). The entertaining nature of games can support the motivation to learn (Garris et al. 2017).

People make decisions and act based on their mental models, norms, and motivations. It is useful to distinguish between decision making and behavior. Decisions can be made habitually or deliberately (Kahneman 2003). The vast majority of decisions are made habitually based on default mental models requiring little effort (Kahneman 2003, Lerner et al. 2015). However, many interventions assume deliberation (den Haan and van der Voort 2018). One goal of experiential learning is to activate deliberate decision making (Flood et al. 2018) and create a moment of recognition that hopefully persists and overrides problematic heuristics in the RLAS.

The learning in the ELAS aims to eventually change real-life behavior in participants and other community members (den Haan and van der Voort 2018), which in the context of commons management, is associated with appropriation, resource provisioning, and institutional service-provisioning actions (Ostrom 1990, Hinkel et al. 2015). Although playing games may have the strongest experiential learning effect on individuals who play or directly witness the game, creating spillover effects to those who did not participate represents a crucial component of the learning process. Community debriefings that include participants beyond those who played the game are especially important for this collective learning and may lay the foundation for future institutional service provision in the RLAS.

IDENTIFYING DESIGN ELEMENTS

Our second objective is the identification of game design features as elements of the ELAS context. Influencing mental models and norms, stimulating motivation, and activating deliberation in the ELAS strongly depend on the learning context. We categorize three context dimensions: player attributes, game rules, and game narrative and experience. Table 1 presents the key features of the three context domains that have been argued to influence learning and behavioral determinants. We describe how each feature influences processes in our conceptual framework.

BACKGROUND ON GAME CASES

The three learning cases we analyze and use to reflect on game intervention cases each included a common pool resource game and community debriefing sessions in which general observations and implications of the game for the real-life management of that resource were discussed. Figure 2 illustrates the study locations. The main game rules are described below and summarized in Fig. 3. More detailed game descriptions are provided in Appendix 1. The game manuals are available on the website at https://gamesforsustainability.org/. The groundwater game simulated the connection between crop choice and groundwater levels. The game was first piloted in 17 communities in rural Andhra Pradesh, where it was conducted twice in the same communities in 2013 and 2014 and compared to 9 control communities (Meinzen-Dick et al. 2016, 2018). It was later applied in 184 communities in Rajasthan, Madhya Pradesh, and Andhra Pradesh. In each community, separate game sessions were conducted, one for men and one for women, with five players in each group. Each group began with a certain groundwater level. In each game round,

Table 1. Key features of the experiential learning context.

Design features	Explanation	Related sources
Game narrative and experience		
Theory of change	The experiential learning action situation (ELAS) should be designed on the basis of a theory around a significant real-life action situation in which current actions lead to undesirable outcomes, which players can potentially address. The issue and response can be abstract or more specific.	Weber et al. 2004, Speelman et al. 2018, Muhamad and Kim 2020
Embedded or isolated intervention	Experiential learning interventions may address important but not all factors influencing behavioral change. A comprehensive theory of change can indicate how the ELAS, in combination with other interventions, can support change. It has been shown that repeated ELAS interventions support learning. Evidence also shows that games are most effective if they are combined with other capacity-development methods.	Weber et al. 2004, Wouters et al. 2013, Scholz et al. 2014, Flood et al. 2018, Muhamad and Kim 2020
Story framing	It has been shown that learning is most effective when it is contextual, and the ELAS structure and possible choices closely reflect the context and key management challenges in the real-life action situation (RLAS). In contrast, simple and more abstract games can activate transcendental values, and their transfer to a specific context could happen at a later step. Subtle manipulations of the narrative can influence learning in desirable or undesirable ways. The intervention can lose credibility or lead to counterproductive learning if the rules of the game are not congruent with the real-life reference of the narrative.	Weber et al. 2004, Wouters et al. 2013, Starks 2014, Medema et al. 2016, Kenter et al. 2016b, Muhamad and Kim 2020, Garcia et al. 2022
Goal definition	Games create an experience structured around goals. The nature of the goal influences behavior and learning in the ELAS. Evidence exists that cooperation is enhanced when games are framed as ethical rather than income-generating challenges. Incentivized payments and common practice in economic behavioral experiments may emphasize material goals, but can also influence learning. Activating relevant transcendental values may support the formation of contextual norms. Ideally, ELAS outcomes mimic RLAS outcomes that are relevant to actors' motivations.	Liberman et al. 2004, Parks et al. 2013, Mørkbak et al. 2014, Starks 2014, Kenter et al. 2016b, Flood et al. 2018, Meinzen-Dick et al. 2018, Speelman et al. 2018, Bartels et al. 2022, Garcia et al. 2022
Facilitation	The more complex a game, the more skilled and experienced facilitation is required. To respond to self-determination needs, facilitation needs to offer spaces for reflection and transformative dialogue rather than teaching solutions. Together with fair game rules, facilitation can create a collective choice system that players perceive as legitimate. It needs to be respectful of the knowledge of all involved, especially marginalized groups. This experience may affect how players want the collective choice rules in the RLAS to be revised. Facilitation also affects the enjoyment of the process.	Scholz et al. 2014, Kenter et al. 2016b, Flood et al. 2018, DeCaro et al. 2021
Game flow	Players are more likely to learn if they feel engaged in the ELAS.	Starks 2014, Muhamad and Kim 2020, Garcia et al. 2022
Graphics of posters and game materials	Graphical material used to explain and facilitate the game is especially important when working with target audiences with limited literacy. Its quality activates visual intelligences as a source of creativity, affecting learning.	Gardner 2000, Starks 2014
Language	Learning is influenced not only by the information content but also by how well the language used is attuned to the audience. Language further activates linguistic intelligences as another source of creativity, affecting learning.	Gardner 2000, Starks 2014, Flood et al. 2018
Integration of any embodiment	People subconsciously react to other bodies in their environment, influencing feelings of connectedness with others and eventually, cooperation. This aspect activates body/kinaesthetic intelligences as another source of creativity, affecting learning.	Gardner 2000, Marsh et al. 2009, Starks 2014
Game length	Often, long games are perceived to be exhausting. Shorter games can capture less complexity and give participants fewer opportunities to learn about the system.	Flood et al. 2018, Garcia et al. 2022
Group competition	Letting more than one group play simultaneously, and allowing them to observe each other can trigger competition between groups, which often strengthens within-group cooperation.	Darwin 1871, Parks et al. 2013, Muhamad and Kim 2020
Game rules		
Response options	Response options need to be aligned to the theory of change. They can be more general or represent a specific action as experienced in a RLAS. Allowing players to set their own rules in the game can activate creativity and satisfy self-determination needs. This can trigger rule acceptance and support a feeling of self-efficacy.	Weber et al. 2004, Ostrom 2009, Flood et al. 2018, Speelman et al. 2018, DeCaro et al. 2021
Payoff structure	The payoffs of a game can illustrate a social dilemma with simple action-outcome relations. Alternatively, they can represent more complex RLAS system dynamics. Because this affects the type of learning, payoffs should be well aligned to the theory of change, the game narrative, and the response options. Payoffs should represent the real-world appropriateness of actions in the ELAS.	Weber et al. 2004, Flood et al. 2018, Speelman et al. 2018, Muhamad and Kim 2020, Garcia et al. 2022
Uncertain payoffs	Payoffs can be fixed or random, representing unpredictable RLAS dynamics.	Weber et al. 2004, Parks et al. 2013, Flood et al. 2018
Sequence of decisions	Decisions can be made simultaneously or sequentially. Sequential decisions express asymmetric power.	Weber et al. 2004, Janssen et al. 2011a, Parks et al. 2013
Repetition of decision	Games can have only one round or repeated rounds. Repeated game rounds allow participants to experiment and revise strategies. Thereby, the decision situation of the next round can depend on the game dynamic in the previous round, which can support learning about long-term path dependencies in the RLAS. In addition, it has been shown that cooperation is more likely in repeated interaction.	Ostrom 2009, Parks et al. 2013, Wouters et al. 2013, Flood et al. 2018, Garcia et al. 2022
Communication	The possibility to interact improves cooperation through multiple mechanisms. It activates people's deeply rooted impulses of curiosity, to wonder, to know, to explain, and to understand. In addition, communication activates interpersonal intelligences as a source of human creativity. Social interaction is the most powerful driver of social learning.	Ostrom and Walker 1991, Gardner 2000, Balliet 2010, Starks 2014, Kenter et al. 2016b, Flood et al. 2018, DeCaro et al. 2021
Consequences	Including rewards or punishments influences cooperation. Consequences are also an important aspect of institutions and including them in the ELAS may encourage institutional change in the RLAS.	Ostrom 2009, Falk et al. 2012, Parks et al. 2013
Encouragement	Feedback on individual and group success or failure supports learning. Feedback can be given individually and secretly or openly for all players to see each other's decisions and outcomes.	Ostrom 2009, Wouters et al. 2013, Starks 2014, Muhamad and Kim 2020

(con'd)

Power	Many learning games give all players the same power in the ELAS. Where power is an issue in the RLAS, unequal endowments, information, or sequential decisions can be integrated to explore how power dynamics influence cooperation.	Parks et al. 2013
Player attributes		
Personality and identity	The combination of individuals with their socioeconomic attributes such as gender, age, education, and skills (for example, to participate in deliberation and negotiation), as well as their role and power in the community, affect social dynamics in the game and the learning. On one hand, games can support empowerment of marginalized groups. On the other hand, it has been argued that to create impact in real-life, games should be played with the actors who have the power to make decisions.	Weber et al. 2004, Scholz et al. 2014, Kenter et al. 2016b, Speelman et al. 2018, Garcia et al. 2022
History/experience with management challenge	Players' past experience with the RLAS affects their behavior in the ELAS and their learning.	Weber et al. 2004, Ostrom 2009, Muhamad and Kim 2020
Group size	Facilitation and the emergence of cooperation are more difficult in larger groups.	Hamburger et al. 1975, Ostrom 1990, Weber et al. 2004

players were asked to choose between a more profitable but more water-intensive crop and a less profitable but more water-efficient crop. A fixed water recharge amount was given after each round. If too many players chose the water-intensive crop, water use would exceed recharge and the groundwater level fell. If the water table went below a threshold, the game was over. The game had a phase without communication and a phase allowing it. The communities were monitored afterward in terms of changes in their rule formulation.

In the surface water game, players had to jointly invest in a virtual dam. In 2016, the game was played in 30 randomly selected villages in Rajasthan (Falk et al. 2019). In each game round, players decided simultaneously to invest a share of their endowment in dam maintenance. The total of all individual contributions determined the group earnings based on a nonlinear payoff function similar to the one in the irrigation games by Cardenas and Carpenter (2008) and Janssen et al. (2011b). The group earnings were distributed equally among players. In 2017, a crop choice decision was added for a second rollout in 60 communities in Madhya Pradesh. In each community, one session was held with two groups of seven players each, including both men and women. In this version, the group investment in dam maintenance determined the amount of irrigation water available in a round. This water could be used to grow a more profitable but more water-consuming crop and a less profitable but more water-efficient crop. All players could only grow a crop and receive game income if the dam maintenance investment was sufficient, and all players chose to grow the water-efficient crop. The game had a phase without communication and a phase allowing it. Game implementation was complemented by a baseline and a follow-up survey of the participating communities as well as 30 control communities to assess the impact of the games (Bartels et al. 2022).

The forestry game, adapted from Janssen et al. (2013), was conducted in 60 habitations in Andhra Pradesh and Rajasthan between 2017 and 2018 (Zhang et al. 2021). In each habitation, two game sessions were carried out, one for men and one for women, with five players in each group. In each round, participants could harvest trees from a shared forest, which gave them income in the game. For every 10 trees remaining in the shared forest, 1 tree was added at the end of the round as regrowth. The maximum number of trees that could be harvested in each round depended on the size of the shared forest at the beginning of that round. Within each game session, participants played three phases in a fixed order: noncommunication, communication, and optional election of monitoring and sanctioning rules. We tested whether alternative graphics of the game board influenced learning.

To test whether individual incentivized payments affected players' choices and learning, all three games randomly allocated half of the sites with individual payments based on their "earnings" in the game, and the other half with a single lump-sum payment to the community.

Key characteristics of game impacts

In 2013 and 2014, the first groundwater pilot in Andhra Pradesh found impacts on mental models and rules. Comparing members of 17 treated communities with randomly selected members of 9 control communities, 1 year after the games were played, respondents in treated communities were significantly less likely to report the widespread belief that crop choices were individual decisions and in principle should not be restricted in any way than respondents in control communities. Treated communities more frequently expressed the need for farmers to cooperate and were also more likely to have introduced water registers or groundwater management-related rules compared to control sites (Meinzen-Dick et al. 2018).

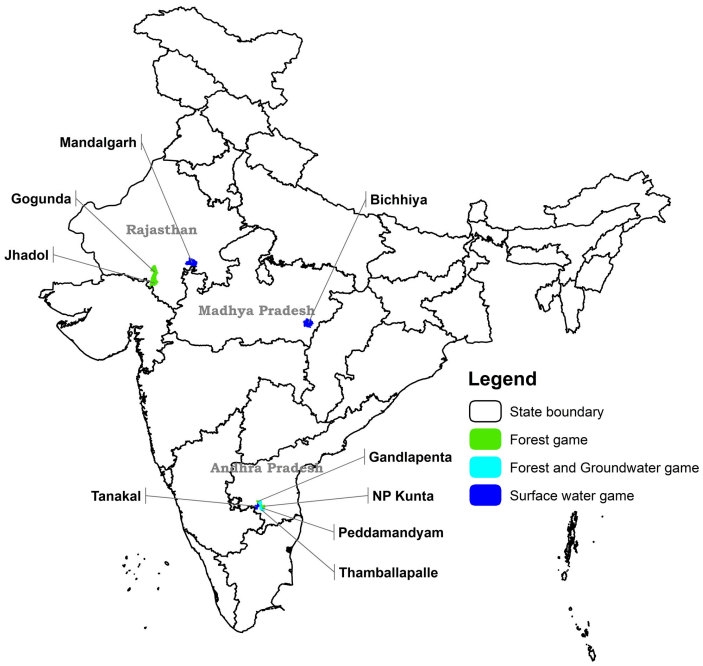
Impacts from the surface water games were assessed in Madhya Pradesh. A comparison of baseline and endline assessments of 60 randomly selected treated and 30 control villages showed that 2 years after the game was played, treated communities reported more dam maintenance activities than control communities (Bartels et al. 2022).

Based on mental model surveys administered immediately after playing the game, participants from the forestry game related the game to real-life aspects such as forest benefits, forest management rules and their enforcement, own provisioning actions, and the importance of trust and collective action. About 1.5 years after the game, we revisited a stratified random sample of 16 out of 60 habitations. Despite the generally positive memory of the game, actual reported changes were limited (Zhang et al. 2021).

RESULTS: REFLECTION ON CASES

Our third objective was to reflect on game features we considered particularly important for influencing behavioral drivers in commons management in three intervention cases from India. We follow the three domains of the ELAS context defined in the framework section. Tables A1a-c in Appendix 1 offer a systematic response to each of the key features of experiential learning context for all three games.

Fig. 2. Case study locations. Source: Census of India, 2011.



Game narrative and experience

The theory of change for all games was built around real-life social dilemmas in which incentives for individual actions conflict with group interests. We expected that participants' positive experience in the ELAS would motivate cooperation in the RLAS.

To illustrate the social dilemmas, all our games had a simplified framing featuring the RLAS. This made it easy for participants to understand the link between actions and consequences. In the RLAS, these actions have delayed effects at large spatial scales, which made it difficult for actors to see causal relations. Imitating real-life management challenges in the ELAS helped players relate to the game. The debriefings indicated that this allowed participants to transfer learning from the ELAS to the RLAS.

The groundwater game created an awareness of links between individual crop choices, local medium-term hydrological dynamics, and income of farmers. Participants in the forestry game related to the framing of a degraded forest condition, the message about the benefits of the forest, and the importance of protecting and sustaining it resonated with community members. The surface water game reflected the challenges of local communities to maintain common water infrastructure and to distribute the rainwater harvested in these structures. This was relevant in most communities in Madhya Pradesh where the game was played. By accident, the game was played in some communities that do not use surface water for irrigation. Attention should be paid to the right fit between tool and RLAS context.

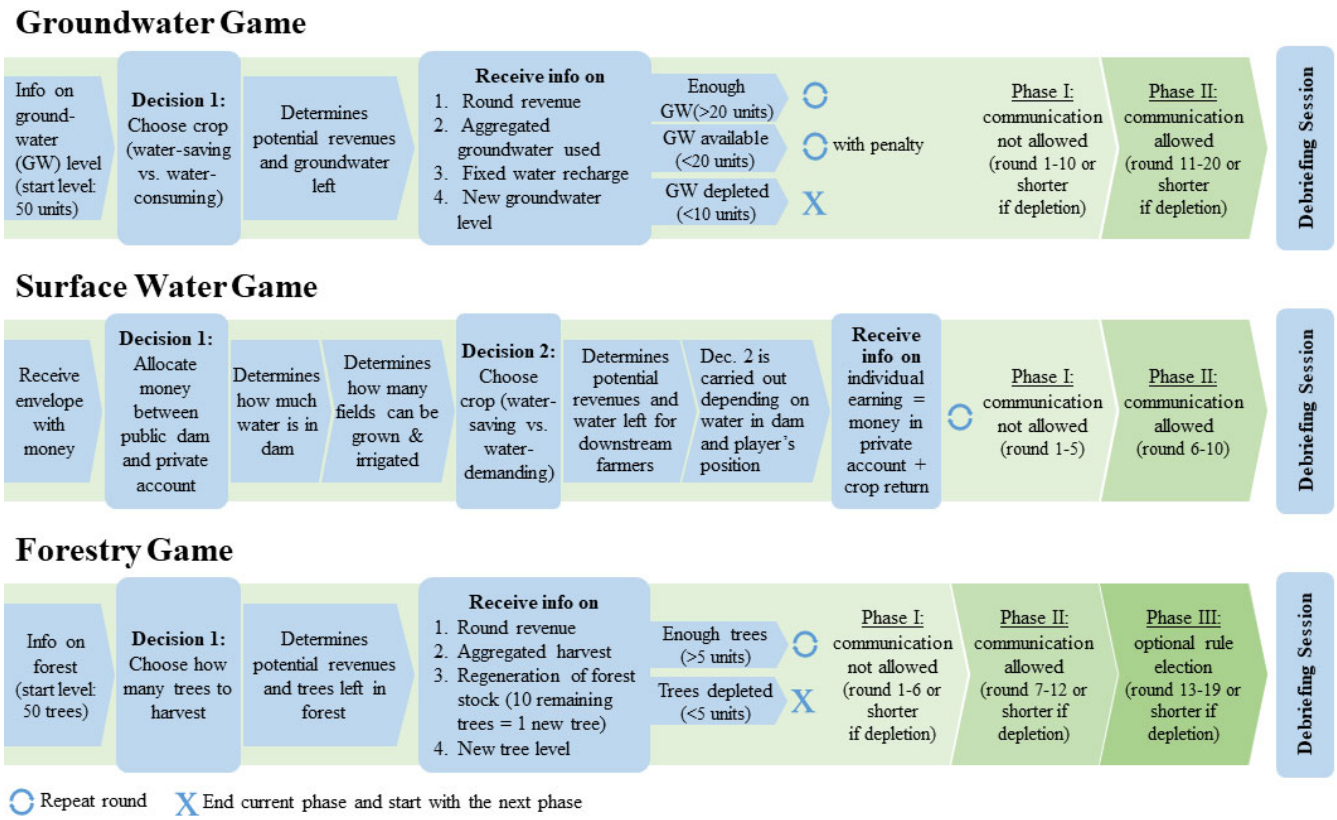
Game boards with different colors corresponding to different levels of ecosystem services provided by varying sizes of the forest were used in different sessions of the forestry game. Because men and women in Rajasthan and Andhra Pradesh responded differently to this treatment, we found it difficult to interpret the inconsistent results. Male groups in Rajasthan showed a more cooperative behavior in the game when playing with a multicolor board. In contrast, female groups in Rajasthan and both gender groups in Andhra Pradesh showed less pro-social behavior when playing with the same board.

Social dilemmas always have potential for conflict. We observed heated discussions in some of our sessions. Some players behaved very opportunistically in the game and convincingly expressed after the session that they wanted to show fellow villagers a mirror of what they believed was happening in their community. Although this was intended, such situations required sensitive facilitation and debriefing to harness "creative tension." It was important that our local facilitators were well grounded in the communities and had a good understanding of the background of potential conflicts to avoid unintended negative effects.

Even the moderate rollouts of the games would not have been easily possible had we relied on experienced professional facilitators. Over multiple days, we developed the community facilitators' capacity to lead through the games. In addition, we reduced the complexity of the game rules to adapt to the capacity of facilitators and players. Development of a mobile application that guided facilitators through the different steps of the surface water game also reduced the demand for facilitation capacity; an android app has now also been developed for the groundwater game.

We experienced, both in the discussion during the games and the debriefing sessions, the temptation from facilitators to hint or give advice. Although this is a way to include expert knowledge in the discussion, it undermines self-determination and may push participants to make decisions they do not really support. When

Fig. 3. Main rules of case study games.



training facilitators, we emphasized the need to be patient and let participants find their own solutions. In the surface water game, we added open questions based on theoretical concepts of commons' management to make participants think and discuss critical aspects of the problems.

The debriefings were critical for the larger community to discuss how the experiences in the game related to their resource use and condition, and the factors influencing it. Debriefings encouraged people to share their perceptions about current and past rules and regulations. But most importantly, debriefings helped communities develop a vision for what should change in the future and what they could contribute to move toward this vision, for example, changes to the rules. The combination of playing the games over multiple rounds and encouraging discussions provoked deliberation processes. Although baseline assessments in our study indicated there were habits of insufficient efforts to manage resources sustainably, the players connected the game to their real-life experiences and discussed the need to change behavior. Players in all the water games expressed that the game helped them better understand how their actions affected other community members and motivated them to deliberate on alternative management. The debriefing sessions also revealed if the game was not addressing a critical management challenge in the specific context, such as fluctuations in water availability or crop prices.

The length of the game was another important aspect in our interventions. The games were supposed to be long enough to enable the benefits of cooperation and to create a memorable experience, but short enough to respect participants' time constraints. We especially noticed in the groundwater game that women were particularly time constrained and, in some cases, depleted the groundwater faster so they could end the game. Finding appropriate times for women to play the game reduced this problem.

We experienced that the games had more positive effects if they complemented other interventions. The longer an NGO had been operating in a community, the more likely the groundwater game was to effect change in behavior. The surface water game was more likely to lead to increased maintenance activities if it was implemented in a site where FES was implementing additional activities.

Game rules

The partners faced the challenge of developing game rules that featured important RLAS social dilemmas and were simple to facilitate in a reasonable time. The groundwater game used two generic crop/income combinations to illustrate that short-term gains can come at long-term and social costs; game payoffs did not reflect the full range of real-life crop income and water requirements. In contrast, the surface water game had very context-specific game rules, with payoffs simulating typical values of the region based on focus group discussions and expert advice.

Although the forestry game fits the real-life biophysical context in terms of forest degradation, we noticed that the choice options did not fit the governance context. The harvesting decision was phrased in terms of cutting trees, but in reality, the government prohibits felling trees. Many communities have their own rules prohibiting tree cutting, even restricting the lopping of branches. Some participants expressed concern that they were being told to cut trees in the game. When communities already had a good understanding of the relationship between cutting trees and forest condition, the simplified rules in the game did not help advance the mental models of players or the community regarding community forest rules.

For all our games, some players questioned the game rules and parameters, requesting more context accuracy to be embedded in the game. In response, in some cases, the framing was adjusted. For instance, many groundwater game participants found the constant water recharge level to be unrealistic because rainfall and recharge fluctuate between years. Thus, subsequent variations of the groundwater game as an intervention allowed for randomizing the amount of recharge based on a roll of the dice.

Flexible designs are a way to help make connections between the ELAS and the RLAS. In the revised groundwater game, participatory design elements were added by offering alternative framings for water extraction decisions and for variable groundwater recharge. In the Rajasthan surface water and the forestry games, game variations such as subsidies, the possibility to sanction, or unequal benefit sharing were optionally played. Depending on what players discussed, the facilitator would introduce one of the variations. This process helped players discuss and understand particular aspects of the management and coordination challenges they felt were most relevant to them.

Our assessments confirm the positive effect of communication in all our games. At the beginning of the games, we prevented players from talking and asked them to make decisions in private. During this game phase, decisions were more erratic and improvements in group earnings were much slower than when discussion was permitted in later rounds. Both players, bystanders, and facilitators enjoyed our games more when communication was allowed. Nevertheless, preventing communication in the first phase offered an opportunity for players to see how others behaved if they were not observed and if the group could not coordinate. This experience underscored the value of communication when it was then allowed.

To support a positive learning experience and make the games more enjoyable, we let two groups play the surface water game simultaneously. Often a friendly competition emerged as the two groups could observe each other, which seemed to strengthen within-group cooperation.

All our games were played over multiple rounds, giving players immediate feedback in a compressed time. This contributed to the players' increased awareness of consequences on third parties and ecological processes, such as the effects of cropping choice and individual water abstraction on groundwater levels, as expressed during the debriefings.

We experimented across all games with different modes of payments. As is common practice in experiential economics, at some sites players received performance-based individual payments, which were compared with players in villages receiving a lump-sum payment to be used for any community purposes. Our results are inconclusive. In the Andhra Pradesh groundwater game, no effect of the payment methods could be found. Two years after playing the surface water game, communities that received performance-based individual payments reported more frequent dam maintenance than the villages that received community payments. In the community forestry game, incentivized payments lowered ELAS harvest for both men and women's groups in Andhra Pradesh and for men's groups in Rajasthan.

Attributes of players

Group composition is an important aspect of creating multiplayer learning spaces. In some communities, we recognized that many women would not speak in public in the presence of men. For this reason, we held separate groups for women and men in the groundwater and forestry games. Facilitators observed active participation of women in the female-only sessions. We then asked ourselves how to facilitate more relational learning. In heterogeneous groups, players with different land size, age, education, and gender could share their perspectives. In the groundwater game, it helped to modify the community-level debriefing practice to have small group discussions separately among women and men, and then to ask representatives of those groups to summarize their insights in the larger group discussion. We got the impression that this helped to negotiate power dynamics. As an alternative approach, we played the surface water game sessions with mixed female-male groups. On average, female players only talked half as much as the men during the discussion. At the same time, in almost every session, we observed interaction between men and women.

Relation was also a discussion among the ELAS designers regarding the trade-off between empowerment of sub-groups and game impacts. By the time of the interventions, the decisions featured in the games were mainly made by men in real life. In the forestry game, women were experimenting with monitoring and sanctioning rules, even though they were traditionally not very much involved in such activities. In the water games, women made crop choices and contributed to dam maintenance even though these decisions were also traditionally dominated by men. Influencing more men may have resulted in stronger institutional service provisions, cropping system shifts on larger areas, and more efforts in dam maintenance. It was a conscious decision to balance the two objectives.

DISCUSSION

Across all three domains of the ELAS context, we struggled with the trade-off between simplicity and relatability and context specificity of games. Games that target more general norms or support learning of a more universal mental model (such as that in social dilemmas, cooperation improves group outcomes) do not require a framing strongly linked to the context. The advantage of simple games is that they are typically easier to facilitate, simple to understand for the players, fast to play (Flood et al. 2018), adapt into modules or building blocks of games, and apply in heterogeneous geographic situations (Speelman et al. 2018). Therefore, simple games can more easily be scaled given the limited number of strong facilitators (Kolavalli and Kerr 2002).

In contrast, more complex and context-specific games may better represent the constraints and opportunities actors face in the RLAS (Garcia et al. 2022). Relevant learning requires that players can relate the ELAS to the relevant RLAS (Medema et al. 2016). This allows game designers to target more specific mental models that relate to causal links between own actions and outcomes at different temporal and spatial scales. Especially in the context of wicked commons management challenges, the major difficulty is often not the biophysical complexity, but the social system complexity, with actors' conflicting agendas, power relations, and beliefs (Pahl-Wostl et al. 2008, Schlüter et al. 2017, Speelman et al. 2018). If players respond to this social system complexity by agreeing on a specific rule in the ELAS, it becomes easier to transfer this rule into real life. New rules can thereby gain additional legitimacy, which later reduces enforcement costs (Falk et al. 2012). Context-specific games can support the identification of locally adapted solutions and improve the participants' feelings of ownership of the solution. When players encounter positive impacts of simulated rules in the game, this experience can strengthen their confidence in the respective institution as well as their capacity, willingness, and feeling of responsibility to invest in the provision of institutional services (Pahl-Wostl et al. 2007, Wigboldus et al. 2016). But as the forestry game shows, context-specific games require detailed understanding of the local context.

Even for more context-specific games, the game rules need to be challenging enough to engage participants but simple enough to allow them to succeed in cooperating. Positive game experiences can create trust required for fruitful collaboration (Hertzog et al. 2014, Meinzen-Dick et al. 2016). Conversely, negative game experiences can support undesirable habits and do harm. Pro-social experiences are therefore important in experiential learning games. It is a challenge to design games in a way that steers players toward more desirable behavior while also allowing them to make their own choices and find their own solutions (Sunstein 2018). Overall, the majority of groups in our surface water game sessions improved their cooperation over the rounds and most managed to reach a cooperative group outcome at the end of the game (Bartels et al. 2022).

Acknowledging the importance of real-life context for collective action and deliberation in the ELAS points to limitations of the experiential learning approach. Changes in mental models and especially shared norms will be strongly influenced by the quality of deliberation affected by local power dynamics, deliberation, and negotiation skills, which are out of the control of the experiential learning facilitators (Scholz et al. 2014, Kenter et al. 2016a). It is critical to carefully frame social interaction in the ELAS through conscious choices in the game rules and attributes of players. Depending on how the game rules are defined, real-life power constellations may be less prominent in the ELAS, which can help bring the voices of different subgroups to the table. A game can surface hidden smoldering conflicts (Shelton et al. 2018) and as a result contribute to questioning current practices. Games provide an informal space in which actors can leave entrenched positions (Pahl-Wostl et al. 2007). Power-sensitive facilitation and game design can reduce the risk of creating destructive conflicts. Debriefings are especially important when strong positive or negative feelings are generated in the game, which would especially apply when conflicts surface (Crookall 2014).

Unlocking a games' potential to trigger participatory institutional change requires nonprescriptive facilitation. Facilitators who can flexibly react to group dynamics can support creation of a constructive group learning exercise (Flood et al. 2018). This requires special capacity-development efforts, especially when working with less experienced facilitators, often used to more conventional teaching (Meinzen-Dick et al. 2018).

Our results confirm that experiential learning more likely leads to intended behavioral change if it is embedded in repeated interactions and complementary interventions (Kolavalli and Kerr 2002, Woodhill 2010, Wouters et al. 2013, Meinzen-Dick et al. 2018). The scaling challenge is therefore to develop effective and efficient intervention packages, potentially containing games as one element. Experiential learning tools such as the ones discussed, can also be incorporated into long-term system change processes in an iterative and adaptive manner (Herrmann et al. 2021). This will intensify the learning experience, help relate it to specific real-life decisions, and support feedback into real-life context. Combining the conceptual thinking of scientists with the applied and context knowledge of NGOs, government partners, and community members was essential in the development of all three described games. The integration of various bodies of knowledge, perspectives, and approaches coproduces socially robust, context-specific, relevant, and holistic solutions, also in the context of game design (Aeberhard and Rist 2009, Jahn et al. 2012, Muhamad and Kim 2020). Codesign processes such as adaptation of the water games, create a feeling of ownership, increase trust, and stimulate commitment among facilitators and other change agents. Codesign increases the legitimacy and impact of the research for development outcomes. Further, it ensures that the time, experience, and resource constraints of facilitators are considered. Eventually, codesign processes increase the likelihood for the solution to be later adopted (Bracken et al. 2015). At the same time, codesign processes are challenging. It is important for all involved parties to move beyond their individual agendas and comfort zones.

CONCLUSIONS

Games have potential as intervention tools for sustainable resource management, but to achieve that potential, careful attention to underlying assumptions in using games is urgently needed in research, policy, and design of interventions. Although it is beyond our scope to synthesize and integrate the massive research on human behavior in multiple disciplines, the reflections on the three cases using the structure of our framework helped us understand the most important features of our games, which we believe have broader theoretical and practical implications.

First, making transparent assumptions in theories of change allows for questioning them and can help identify effective ways to influence behavioral change. It follows from our conceptual framework that experiential learning can mainly influence mental models and norms. Consequently, they are unlikely to change real-life behavior if mental models and norms are not the main constraints for desirable behavior. Making assumptions transparent can help to understand why or under what conditions experiential learning works or fails (Davis et al. 2015). Acknowledging the diversity of intervention tools, we conclude that there is a critical need for more systematic choices of the right tools for the right purpose.

Second, participants from the experiential learning must be able to relate the game to their real lives. Game designers should consciously decide whether they want to influence more universal mental models or mental models related to a more specific management challenge. The game narrative and rules must fit the specific resource management context if interventions are to influence a particular behavior. A social dilemma in the game should, at least in its basic form, represent a real-life dilemma. Too often communities experience interventions that are only perceived as important by outsiders, not considering the specific context.

Third, allowing communication for a considerable part of the game is necessary to support social learning. Communication allows players to coordinate and, thus, have a positive cooperation experience. The latter is required to ensure that groups align their mental models regarding the benefits of coordination and cooperation. Game designers should be conscious that how they structure the ELAS exercises power over the learning process. A tension exists between giving players freedom to deliberate without external interference and the intent to support learning around a specific real-life issue through process design and active facilitation (Kenter et al. 2016a, b).

Fourth, games benefit greatly from the subsequent debriefing that can serve as a bridge to transfer the lessons from the games to real life. In the debriefing, it is important that facilitators allow space for participants to find their own solutions. This will increase the acceptance of the solutions as well as help to better adapt the solutions to the specific context of the session's participants. As a result, the experiential learning will more likely lead to actual behavioral change. If the solutions found are manifested in local rules, this bottom-up process is likely to support better-adapted resource governance.

The conceptual framework we presented can provide a structure for future impact assessments of experiential learning. We propose a stepwise approach including: (1) assessing indicators for changes in mental models and descriptive and injunctive norms, (2) measuring real-life behavioral changes, and (3) assessing changes in real-life system contexts.

Our conceptualization and reflection on experiential learning shed light on future research needs, including the following:

1. Better understanding of the role of people's motivations in social dilemma situations and in experiential learning interventions is needed. How would alternative goal definitions affect the lessons participants learn from playing the game? In particular, what is the role of individual incentivized payments?
2. Many commercial multiplayer games include an element of competition between groups. It has been shown that such competition can strengthen cooperation within groups. Intergroup competition may have harmful effects in real life but can support cooperation on a community level. The interaction between competition and cooperation at different scales of social structures deserves additional research.
3. More research is required to understand the effect of visual design, language, and embodiment elements on behavioral responses and learning for people with different socioeconomic attributes. Advanced cognitive testing methods may be required to better understand the role of specific design features.
4. The potential application of experiential learning games in situations in which power rather than mental models and norms drive commons management needs to be better understood. We see the potential to empower marginalized groups in the game sessions but recommend studying the impact of game dynamics on real-life power dynamics.
5. Gender dynamics deserve particular attention to (1) identify the conditions under which women can participate actively in mixed male-female games and (2) to identify how social learning among women and men relates to their gender-differentiated roles in resource management.
6. The desired impacts of our game interventions require behavioral change at both the individual and community levels. This indicates the importance of complementary interventions such as debriefing sessions. More thought is required on how experiential learning experienced directly by (a limited number of) participants can be diffused effectively to the wider community, triggering sustainable commons management on a larger scale.

Acknowledgments:

This work was undertaken as part of the CGIAR Research Program on policies, institutions, and market as well as the program on water, land, and ecosystems. This work received financial support from the German Federal Ministry for Economic Cooperation and Development, commissioned and administered through the Deutsche Gesellschaft für Internationale Zusammenarbeit, grant number: 81250397. It was further supported by the German Federal Ministry of Education and Research, grant number: 01UC2114A. We have benefited greatly from discussions with Marco Janssen, Bjorn Vollan, and Juan Camilo Cárdenas over several years as we developed and implemented the games. We thank Mahesh Jadav for editing the map of the project sites and Lee Dixon for designing a figure, which captures the complexity of our conceptual thinking in an accessible and attractive way. Special acknowledgment goes to all Foundation for Ecological Security staff and participating communities for their time, assistance, and hospitality. The opinions expressed here belong to the authors, and do not necessarily reflect those of the organizations involved.

Data Availability:

Links to the data are included in Appendix 1, which describe the different game tools.

LITERATURE CITED

Aeberhard, A., and S. Rist. 2009. Transdisciplinary co-production of knowledge in the development of organic agriculture in Switzerland. Ecological Economics 68 (4):1171-1181. https://doi.org/10.1016/j.ecolecon.2008.08.008

Baird, J., R. Plummer, C. Haug, and D. Huitema. 2014. Learning effects of interactive decision-making processes for climate change adaptation. Global Environmental Change 27:51-63. https://doi.org/10.1016/j.gloenvcha.2014.04.019

Balliet, D. 2010. Communication and cooperation in social dilemmas: a meta-analytic review. Journal of Conflict Resolution 54(1):39-57. https://doi.org/10.1177/0022002709352443

Barreteau, O., F. Bousquet, and J. M. Attonaty. 2001. Role-playing games for opening the black box of multi-agent systems: method and lessons of its application to Senegal River Valley irrigated systems. Journal of Artificial Societies and Social Simulation 4(2):1-5. https://www.jasss.org/4/2/5.html

Bartels, L., T. Falk, V. Duche, and B. Vollan. 2022. Experimental games in transdisciplinary research: the potential importance of individual payments. Journal of Environmental Economics and Management 113(11):102631. https://doi.org/10.1016/j.jeem.2022.102631

Becu, N., M. Amalric, B. Anselme, E. Beck, X. Bertin, E. Delay, N. Long, N. Marilleau, C. Pignon-Mussaud, and F. Rousseaux. 2017. Participatory simulation to foster social learning on coastal flooding prevention. Environmental Modelling and Software 98:1-11. https://doi.org/10.1016/j.envsoft.2017.09.003

Bousquet, F., O. Barreteau, P. d'Aquino, M. Etienne, S. Boissau, S. Aubert, C. Le Page, D. Babin, and J.-C. Castella. 2002. Multi-agent systems and role games: collective learning processes for ecosystem management. Pages 248-285 in M. A. Janssen, editor. Complexity and ecosystem management: the theory and practice of multi-agent systems. E. Elgar, Cheltenham, UK. https://doi.org/10.4337/9781781957240.00021

Bracken, L. J., H. A. Bulkeley, and G. Whitman. 2015. Transdisciplinary research: understanding the stakeholder perspective. Journal of Environmental Planning and Management 58(7):1291-1308. https://doi.org/10.1080/0964056-8.2014.921596

Cardenas, J. C., and J. Carpenter. 2008. Behavioural development economics: lessons from field labs in the developing world. Journal of Development Studies 44(3):311-338. https://doi.org/10.1080/00220380701848327

Cialdini, R. B. 2003. Crafting normative messages to protect the environment. Current Directions in Psychological Science 12 (4):105-109. https://www2.psych.ubc.ca/~schaller/Psyc591Readings/Cialdini2003.pdf

Craven, J., H. Angarita, G. A. Corzo Perez, and D. Vasquez. 2017. Development and testing of a river basin management simulation game for integrated management of the Magdalena-Cauca River basin. Environmental Modelling and Software 90:78-88. https://doi.org/10.1016/j.envsoft.2017.01.002

Crookall, D. 2014. Engaging (in) gameplay and (in) debriefing. Simulation and Gaming 45(4-5):416-427. https://doi.org/10.1177/1046878114559879

Darwin, C. 1871. The descent of man and selection in relation to sex. 2003 edition. Gibson Square Books, London, UK. https://doi.org/10.5962/bhl.title.2092

Davis, R., R. Campbell, Z. Hildon, L. Hobbs, and S. Michie. 2015. Theories of behaviour and behaviour change across the social and behavioural sciences: a scoping review. Health Psychology Review 9(3):323-344. https://doi.org/10.1080/17437199.2014.941722

DeCaro, D. A., M. A. Janssen, and A. Lee. 2021. Motivational foundations of communication, voluntary cooperation, and self-governance in a common-pool resource dilemma. Current Research in Ecological and Social Psychology 2:100016. https://doi.org/10.1016/j.cresp.2021.100016

den Haan, R.-J., and M. C. van der Voort. 2018. On evaluating social learning outcomes of serious games to collaboratively address sustainability problems: a literature review. Sustainability 10(12):4529. https://doi.org/10.3390/su10124529

Falk, T., S. Kumar, and S. Srigiri. 2019. Experimental games for developing institutional capacity to manage common water infrastructure in India. Agricultural Water Management 221 (219):260-269. https://doi.org/10.1016/j.agwat.2019.05.005

Falk, T., B. Vollan, and M. Kirk. 2012. Analysis of material, social, and moral governance in natural resource management in southern Namibia. International Journal of the Commons 6 (2):271-301. https://doi.org/10.18352/ijc.307

Ferrero, G., F. Bichai, and M. Rusca. 2018. Experiential learning through role-playing: enhancing stakeholder collaboration in water safety plans. Water 10(2):227. https://doi.org/10.3390/w10020227

Flood, S., N. A. Cradock-Henry, P. Blackett, and P. Edwards. 2018. Adaptive and interactive climate futures: systematic review of 'serious games' for engagement and decision-making. Environmental Research Letters 13(6):063005. https://doi.org/10.1088/1748-9326/aac1c6

Garcia, C. A., S. Savilaakso, R. W. Verburg, N. Stoudmann, P. Fernbach, S. A. Sloman, G. D. Peterson, M. B. Araújo, J.-F. Bastin, J. Blaser, L. Boutinot, T. W. Crowther, H. Dessard, A. Dray, S. Francisco, J. Ghazoul, L. Feintrenie, E. Hainzelin, F. Kleinschroth, B. Naimi, I. P. Novotny, J. Oszwald, S. A. Pietsch, F. Quétier, B. E. Robinson, M. Sassen, P. Sist, T. Sunderland, C. Vermeulen, L. Wilmé, S. J. Wilson, F. Zorondo-Rodríguez, and P. O. Waeber. 2022. Strategy games to improve environmental policymaking. Nature Sustainability 5(6):464-471. https://doi.org/10.1038/s41893-022-00881-0

Gardner, H. E. 2000. Intelligence reframed: multiple intelligences for the 21st century. Hachette, New York, New York, USA.

Garris, R., R. Ahlers, and J. E. Driskell. 2017. Games, motivation, and learning: a research and practice model. Pages 441-467 in F. Jentsch, M. Curtis, and E. Salas, editors. Simulation in aviation training. Routledge, England, UK.

Greenhalgh, T., S. Thorne, and K. Malterud. 2018. Time to challenge the spurious hierarchy of systematic over narrative reviews? European Journal of Clinical Investigation 48(6):e12931. https://doi.org/10.1111/eci.12931

Hamburger, H., M. Guyer, and J. Fox. 1975. Group size and cooperation. Journal of Conflict Resolution 19(3):503-531. https://doi.org/10.1177/002200277501900307

Herrmann, D. L., K. Schwarz, C. R. Allen, D. G. Angeler, T. Eason, and A. Garmestani. 2021. Iterative scenarios for social-ecological systems. Ecology and Society 26(4):8. https://doi.org/10.5751/ES-12706-260408

Hertzog, T., J.-C. Poussin, B. Tangara, I. Kouriba, and J.-Y. Jamin. 2014. A role playing game to address future water management issues in a large irrigated system: experience from Mali. Agricultural Water Management 137:1-14. https://doi.org/10.1016/j.agwat.2014.02.003

Hinkel, J., M. E. Cox, M. Schlüter, C. R. Binder, and T. Falk. 2015. A diagnostic procedure for applying the social-ecological systems framework in diverse cases. Ecology and Society 20(1):32. https://doi.org/10.5751/ES-07023-200132

Jahn, T., M. Bergmann, and F. Keil. 2012. Transdisciplinarity: between mainstreaming and marginalization. Ecological Economics 79:1-10. https://doi.org/10.1016/j.ecolecon.2012.04.017

Janssen, M. A., J. M. Anderies, and J.-C. Cardenas. 2011a. Head-enders as stationary bandits in asymmetric commons: comparing irrigation experiments in the laboratory and the field. Ecological Economics 70(9):1590-1598. https://doi.org/10.1016/j.ecolecon.2011.01.006

Janssen, M. A., J. M. Anderies, and S. R. Joshi. 2011b. Coordination and cooperation in asymmetric commons dilemmas. Experimental Economics 14(4):547-566. https://doi.org/10.1007/s10683-011-9281-9

Janssen, M. A., F. Bousquet, J.-C. Cardenas, D. Castillo, and K. Worrapimphong. 2013. Breaking the elected rules in a field experiment on forestry resources. Ecological Economics 90:132-139. https://doi.org/10.1016/j.ecolecon.2013.03.012

Jones, N. A., H. Ross, T. Lynam, P. Perez, and A. Leitch. 2011. Mental models: an interdisciplinary synthesis of theory and methods. Ecology and Society 16(1):46. https://doi.org/10.5751/ES-03802-160146

Kahneman, D. 2003. Maps of bounded rationality: psychology for behavioral economics. American Economic Review 93 (5):1449-1475. https://doi.org/10.1257/000282803322655392

Kenter, J. O., M. S. Reed, and I. Fazey. 2016a. The deliberative value formation model. Ecosystem Services 21:194-207. https://doi.org/10.1016/j.ecoser.2016.09.015

Kenter, J. O., R. Bryce, M. Christie, N. Cooper, N. Hockley, K. N. Irvine, I. Fazey, L. O'Brien, J. Orchard-Webb, N. Ravenscroft, C. M. Raymond, M. S. Reed, P. Tett, and V. Watson. 2016b. Shared values and deliberative valuation: future directions. Ecosystem Services 21:358-371. https://doi.org/10.1016/j.ecoser.2016.10.006

Kolavalli, S., and J. Kerr. 2002. Scaling up participatory watershed development in India. Development and Change 33(2):213-235. https://doi.org/10.1111/1467-7660.00248

Kolb, A. Y., and D. A. Kolb. 2009. Experiential learning theory: a dynamic, holistic approach to management learning, education and development. Pages 42-68 in S. J. Armstrong and C. Fukami, editors. The SAGE handbook of management learning, education and development. Sage Publications, London, UK. https://doi.org/10.4135/9780857021038.n3

Lerner, J. S., Y. Li, P. Valdesolo, and K. S. Kassam. 2015. Emotion and decision making. Annual Review of Psychology 66:799-823. https://doi.org/10.1146/annurev-psych-010213-115043

Liberman, V., S. M. Samuels, and L. Ross. 2004. The name of the game: predictive power of reputations versus situational labels in determining prisoner's dilemma game moves. Personality and Social Psychology Bulletin 30(9):1175-1185. https://doi.org/10.1177/0146167204264004

Marsh, K. L., M. J. Richardson, and R. C. Schmidt. 2009. Social connection through joint action and interpersonal coordination. Topics in Cognitive Science 1(2):320-339. https://doi.org/10.1111/j.1756-8765.2009.01022.x

Maslow, A. H. 1981. Motivation and personality. Prabhat Prakashan Books, New Delhi, India.

Max-Neef, M. A. 1991. Human scale development: conception, application and further reflections. Apex, New York, New York, USA. http://www.wtf.tw/ref/max-neef.pdf

Medema, W., A. Furber, J. Adamowski, Q. Zhou, and I. Mayer. 2016. Exploring the potential impact of serious games on social learning and stakeholder collaborations for transboundary watershed management of the St. Lawrence River basin. Water 8 (5):175. https://doi.org/10.3390/w8050175

Meinzen-Dick, R., R. Chaturvedi, L. Domènech, R. Ghate, M. A. Janssen, N. D. Rollins, and K. Sandeep. 2016. Games for groundwater governance: field experiments in Andhra Pradesh, India. Ecology and Society 21(3):38. https://doi.org/10.5751/ES-08416-210338

Meinzen-Dick, R., M. A. Janssen, S. Kandikuppa, R. Chaturvedi, K. Rao, and S. Theis. 2018. Playing games to save water: collective action games for groundwater management in Andhra Pradesh, India. World Development 107:40-53. https://doi.org/10.1016/j.worlddev.2018.02.006

Meyer, S., P. Santos, and F. Yang. 2021. Economic games can be used to promote cooperation in the field. Proceedings of the National Academy of Sciences 118(47):e2026046118. https://doi.org/10.1073/pnas.2026046118

Mørkbak, M. R., S. B. Olsen, and D. Campbell. 2014. Behavioral implications of providing real incentives in stated choice experiments. Journal of Economic Psychology 45:102-116. https://doi.org/10.1016/j.joep.2014.07.004

Muhamad, J. W., and S. Kim. 2020. Serious games as communicative tools for attitudinal and behavioral change. Pages 141-162 in H. D. O'Hair and M. J. O'Hair, editors. The handbook of applied communication research. Wiley, Hoboken, New Jersey, USA. https://doi.org/10.1002/9781119399926.ch9

Ostrom, E. 1990. Governing the commons: the evolution of institutions for collective action. Cambridge University Press, Cambridge, UK. https://doi.org/10.1017/CBO9780511807763

Ostrom, E. 2005. Understanding institutional diversity. Princeton University Press, Princeton, New Jersey, USA.

Ostrom, E. 2007. A diagnostic approach for going beyond panaceas. Proceedings of the National Academy of Sciences 104 (39):15181-15187. https://doi.org/10.1073/pnas.0702288104

Ostrom, E. 2009. Building trust to solve commons dilemmas: taking small steps to test an evolving theory of collective action. Pages 207-228 in S. A. Levin, editor. Games, groups, and the global good. Springer, Berlin, Germany.

Ostrom, E. 2011. Background on the institutional analysis and development framework. Policy Studies Journal 39(1):7-27. https://doi.org/10.1111/j.1541-0072.2010.00394.x

Ostrom, E., and J. Walker. 1991. Communication in a commons: cooperation without external enforcement. Pages 287-322 in T. R. Palfrey, editor. Laboratory research in political economy. University of Michigan Press, Ann Arbor, Michigan, USA.

Pahl-Wostl, C., M. Craps, A. Dewulf, E. Mostert, D. Tabara, and T. Taillieu. 2007. Social learning and water resources management. Ecology and Society 12(2):5. https://doi.org/10.5751/ES-02037-120205

Pahl-Wostl, C., and M. Hare. 2004. Processes of social learning in integrated resources management. Journal of Community and Applied Social Psychology 14(3):193-206. https://doi.org/10.1002/casp.774

Pahl-Wostl, C., E. Mostert, and D. Tàbara. 2008. The growing importance of social learning in water resources management and sustainability science. Ecology and Society 13(1):24. https://doi.org/10.5751/ES-02352-130124

Parks, C. D., J. Joireman, and P. A. M. Van Lange. 2013. Cooperation, trust, and antagonism: how public goods are promoted. Psychological Science in the Public Interest 14 (3):119-165. https://doi.org/10.1177/1529100612474436

Schlüter, M., A. Baeza, G. Dressler, K. Frank, J. Groeneveld, W. Jager, M. A. Janssen, R. R. McAllister, B. Müller, K. Orach, N. Schwarz, and N. Wijermans. 2017. A framework for mapping and comparing behavioural theories in models of social-ecological systems. Ecological Economics 131:21-35. https://doi.org/10.1016/j.ecolecon.2016.08.008

Scholz, G., A. Dewulf, and C. Pahl-Wostl. 2014. An analytical framework of social learning facilitated by participatory methods. Systemic Practice and Action Research 27(6):575-591. https://doi.org/10.1007/s11213-013-9310-z

Shelton, R. E., M. A. Janssen, and R. Meinzen-Dick. 2018. Measuring learning from interventions through participatory processes. Center for Behavior, Institutions and the Environment (CBIE) working paper series. Arizona State University, Tempe, Arizona, USA. https://complexity.asu.edu/sites/default/files/papers/cbie_wp_2018-004.pdf

Speelman, E. N., M. van Noordwijk, and C. Garcia. 2018. Gaming to better manage complex natural resource landscapes. Pages 1-11 in Co-investment in ecosystem services: global lessons from payment and incentive schemes. World Agroforestry Centre, Nairobi, Kenya.

Starks, K. 2014. Cognitive behavioral game design: a unified model for designing serious games. Frontiers in Psychology 5:28. https://doi.org/10.3389/fpsyg.2014.00028

Sunstein, C. R. 2018. "Better off, as judged by themselves": a comment on evaluating nudges. International Review of Economics 65(1):1-8. https://doi.org/10.1007/s12232-017-0280-9

van Vugt, M., V. Griskevicius, and P. W. Schultz. 2014. Naturally green: harnessing stone age psychological biases to foster environmental behavior. Social Issues and Policy Review 8 (1):1-32. https://doi.org/10.1111/sipr.12000

Weber, J. M., S. Kopelman, and D. M. Messick. 2004. A conceptual review of decision making in social dilemmas: applying a logic of appropriateness. Personality and Social Psychology Review 8(3):281-307. https://doi.org/10.1207/s15327957pspr0803_4

Wigboldus, S., L. Klerkx, C. Leeuwis, M. Schut, S. Muilerman, and H. Jochemsen. 2016. Systemic perspectives on scaling agricultural innovations. A review. Agronomy for Sustainable Development 36(3):46. https://doi.org/10.1007/s13593-016-0380-z

Woodhill, J. 2010. Capacities for institutional innovation: a complexity perspective. IDS Bulletin 41(3):47-59. https://doi.org/10.1111/j.1759-5436.2010.00136.x

Wouters, P., C. van Nimwegen, H. van Oostendorp, and E. D. van der Spek. 2013. A meta-analysis of the cognitive and motivational effects of serious games. Journal of Educational Psychology 105 (2):249-265. https://doi.org/10.1037/a0031311

Zhang, W., R. S. Meinzen-Dick, S. Valappanandi, R. Balakrishna, H. Reddy, M. A. Janssen, L. Thomas, P. Priyadarshini, S. Kandikuppa, R. Chaturvedi, and R. Ghate. 2021. Norms, gender, and payment method affect extraction behavior in a framed field experiment on community forestry in India. International Food Policy Research Institute, Washington, D.C., USA.

Appendix 1: Detailed description of the games

Groundwater Game

This game was designed to address the depletion of groundwater in areas of confined aquifers (such as hard rock areas) where local collective action could be effective. The game targets one of the major factors influencing groundwater consumption: crop choice, particularly during India's rabi season (following the main monsoon season), when there is some rainfall but farmers depend heavily on groundwater.

This problem context was translated into the game context by presenting players with one main decision affecting groundwater consumption: choosing a crop to grow. The different crop choices not only have different water requirements (water-intensive and water-saving) but also yield different private returns (high and low). The groundwater is a publicly available but a finite resource. Players have a short-run incentive to grow a water-intensive crop with higher private returns rather than a water-saving crop with lower returns. However, the water-demanding crop depletes the water table. If too many players decide to grow the water-demanding crop, groundwater falls below the predetermined threshold and the game ends, leaving players without further revenues. Choosing the sustainable crop leads to lower returns but with a stable time horizon securing the livelihood of farmers. Thus, the socially optimal strategy under the assumption that the game is infinite—as the story framing suggested—is for all players to choose the sustainable crop.

The game was played in 17 randomly chosen communities in Andhra Pradesh, with 9 control sites for comparison. At each site, there was one group of five women and one group of five men who played. Other community members were allowed to observe.

At the beginning of the game, players were informed that they were sharing a total of 50 groundwater units. In each round of play, each player was asked to choose privately between crop A that takes 1 unit of water and yields 2 units of (play) money and crop B that takes 3 units of water but yields 5 units of (play) money. Each round was the equivalent of a rabi season. This was explained to players using a large flexi-banner that showed the image of a well with 50 units of water marked on it along with the crop earnings from the water-intensive and water-saving crops. The banner was displayed where all players could see it. The facilitators asked the players about the major crops sown in the region during rabi season, and images of relevant water-intensive and water-saving crops were affixed to the poster.

After each round, the facilitators recorded the individual crop choices, gave each player their respective crop earnings in play money, and calculated the aggregate groundwater use of all players, adding 2 units for domestic water use. The total groundwater use of the group was deducted from the previous groundwater level. After every round, the group received an amount of water recharge. In earlier versions of the game, a default recharge of 7 units was added to the groundwater level. Later, to account for the variability of rainfall, the facilitators invited a player or a member of the audience to cast two dice, and the value obtained was the recharge available for the next round. The final water level at the end of the round was shown on the banner using a movable strip and written in tabular form.

If the remaining groundwater level was above 10 units, a new game round started. If it fell below 20 units, all players were charged 2 units of money for domestic water, which was subtracted from their private earnings. If the groundwater level fell below 10 units, there was not enough groundwater left for

cultivation and the game ended. The players did not know that the game would end after 10 rounds if the water was not depleted earlier.

When the game was introduced, there were two practice rounds to ensure that all players understood how it worked. Then the game was played in two phases of up to 10 rounds per phase (though the number of rounds in each phase was never announced). In phase 1, no communication was allowed. After a maximum of 10 rounds, or when groundwater was depleted, players were allowed to discuss what happened during the rounds and what they would want to do differently. Then phase 2 began, with up to 2 minutes of communication between each round. Again, after a maximum of 10 rounds or when the groundwater fell below 10 units, the entire game ended.

The game sessions were followed by a debriefing for the whole community, in which the facilitator showed actual community trends in groundwater levels, players discussed their experiences in the game, and the community discussed implications for their community.

To test whether incentivized payments affect decision-making, half the sites were randomly selected to give player individual payments based on their winnings in the game; the other half received a lump sum contribution to the community, but players were not given individual payments.

The complete protocol of the game is open accessible at:
https://gamesforsustainability.org/practitioners/#groundwater-game

The data on follow up surveys in 2016 after the game was played in Andhra Pradesh is open accessible under:
https://www.ifpri.org/publication/experimental-games-strengthening-collective-action-learning-field-experiments-india

Table A1.1. Key features of the groundwater game.

Design Features	**Explanation**
Game narrative and experience	
Theory of change	The citizens in the research areas increasingly rely on groundwater resources for irrigation and water-intensive cash crops. Consequently, the groundwater is subject to depletion. Growing water-saving crops is a key means to reduce groundwater use and make water levels more sustainable.
Embedded or isolated intervention	The game was played at sites where the NGOs Foundation for Ecological Security (FES) or Jana Jagriti were already implementing other community mobilization and water management interventions over a long term.
Story framing	The story framing was developed by the research team and FES to reflect the factors affecting groundwater use but standardized across sites, with generic “Crop A” and “Crop B.”
Goal definition	The goal was defined as generating individual income.
Facilitation	Local community members were trained in facilitating the games and debriefings. They were requested to interfere as little as possible in discussions, but provided some background on community groundwater levels at the beginning of the debriefing.
Game flow	In the facilitation training, emphasis was placed on good coordination among the facilitation team. This considerably reduced the time that players

	sat idle.
Graphics of posters and game materials	Illustrations including drawings, which reflected the regional context, were used to explain the game. Understanding the material did not require the ability to read.
Language	During pre-tests, the explanations were repeatedly revised with the local field team to ensure that they are well understood.
Integration of any embodiment	No expression of ideas, quality, or feelings in a tangible or visible form was included int the intervention.
Game length	Each game session took approximately 1.5 hours.
Debriefing	After the games were played with men's and women's groups, there was a communitywide debriefing allowing players and other community members to discuss how the game related to their real-life water management challenges.
Group competition	The game sessions with men and women were played separately, but other community members were allowed to observe the games.
Game rules	
Response options	In line with our theory of change, the players could make one decision: choose between water-intensive and water-saving crops, which directly impacted future groundwater availability and income. In game phase 2, players were allowed to communicate, giving them a chance to set their own rules in the game.
Payoff structure	The payoff structure was simplified to make the trade-offs between income and sustainability clear.
Uncertain payoffs	The initial game version had no random function. In the later one, water recharge was randomized, which affected the payoffs.
Sequence of decisions	Decisions on the crop choices were made simultaneously by all players in each round.
Repetition of decision	Each game session had a maximum of 20 rounds, with two game phases with a maximum of 10 rounds each. Rounds within a phase were interconnected: groundwater consumption of the previous round determined the groundwater level in the next round.
Communication	During game phase 1, players were not allowed to talk. In phase 2, discussions were allowed and encouraged between rounds.
Consequences	Choices had direct consequences for individual payoffs, the group's total payoff, and the groundwater level.
Encouragement	Players received open information on group impact on the water levels, and individual information on their earnings. In phase 2, players could choose to share information on their individual earnings.
Power	Our game did not specifically address power issues; all players were assumed to be irrigating the same amount of land.
Player attributes	
Personality and identity	Groups were divided by gender, but all were to be from households using groundwater for irrigation. Socioeconomic profiles of players varied within men's and women's groups.
History/experience with management challenges	Players come from households that use groundwater for irrigation. Most communities had experienced falling water tables.

Group size	The game was played with two groups of five players each.

Surface Water Game

This game was designed to address poor local collective action in surface water management. Poor management leads to poor groundwater recharge and limited availability of surface water, which is needed for irrigation during the rabi season. The game addresses maintenance of dam structures and farmers' crop choices, which are the major factors influencing water availability and consumption.

This context is translated in the game into two consecutive decisions, similar to other irrigation games, with a provision and an appropriation decision under asymmetric resource access (Janssen et al. 2011a; Cárdenas et al. 2013). First, players have to allocate a given endowment between the maintenance of the public dam and their private savings. The group's total investment in dam maintenance determines the water-holding capacity of the dam in this round (see Fig A1.1, panel A). Second, individual players choose between growing an unsustainable or a sustainable crop. The unsustainable crop requires 5,000 m^3 of water and has a net return of INR 15,000. The sustainable crop requires 3,000 m^3 of water and has a net return of INR 13,000. Thus, players have an incentive to grow a water-intensive crop with higher private returns rather than the water-saving crop with lower returns. However, the number of players who can grow crops depends on the total investment in dam maintenance (decision 1). Under optimal investments, all seven players can grow the water-saving crop. If even one player chooses the water-intensive crop, the available water will not suffice for all players to grow a crop (see Fig A1.1, panel B).

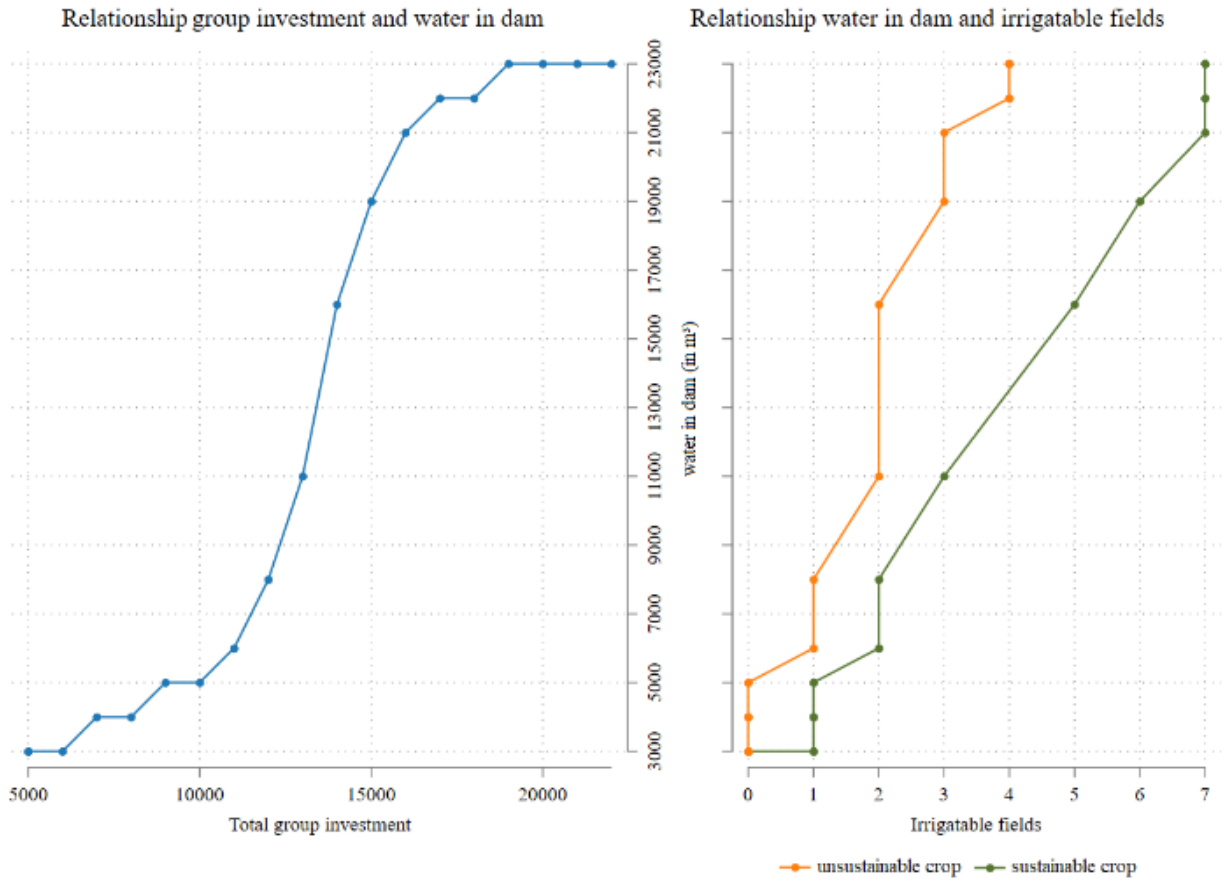


Figure A1.1. Functions of group investment and water availability and of possible crop choices depending on water availability.

Additionally, the players' fields are located sequentially along the feeder channel. Upstream players have preferential access to the water in the dam. For example, how much water player 2 can take for irrigation depends on how much water was provided in step 1 and how much water was left by player 1. Player 3 can only take what was left after player 2 and so forth through player 7. This relationship increases the incentive to free-ride. The socially optimal strategy is for all players to invest about 80 percent of their endowment to maintenance of the public dam and to grow the sustainable crop with lower water requirements and lower returns. This maximizes earnings at the group level (see Figure A1.2). Any heterogeneous crop choice results in lower group income. The difficulty lies in the interdependencies of water availability, crop type, and players' positions and their choices. By interlinking these factors and thus creating complex impact chains, the game mimics the collective action problems around surface water management in the target communities.

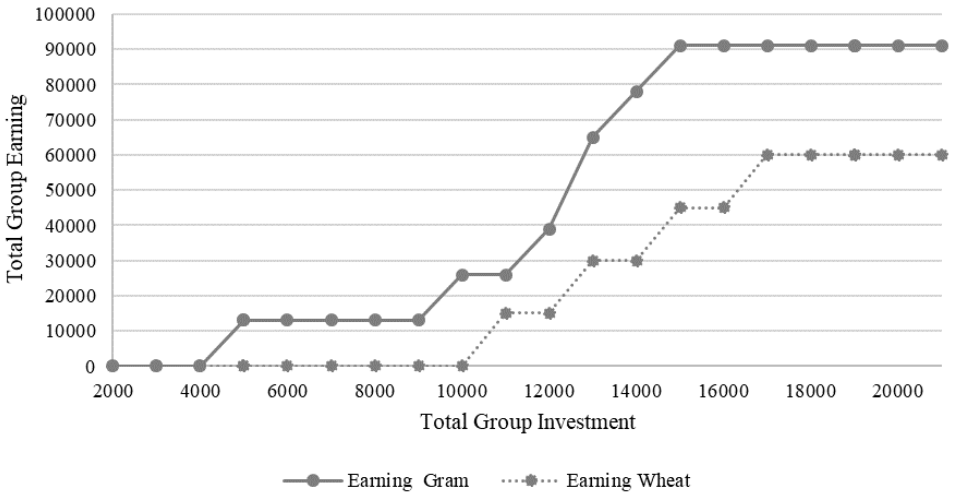


Figure A1.1. Relationship between Total Group Investment and Group Earnings, Based Only on the Net Returns of Crops.

The game was played in 60 randomly chosen sites in Madhya Pradesh. At each site, two groups, consisting of seven players each (men and women), played in parallel. Special attention was paid to including women in the games. Other community members were allowed to observe. Each group played a maximum of 10 rounds. The rounds were divided in 2 phases with 5 rounds in each phase. In phase 1, participants played with full anonymity and without communication. They did not know who was in which group nor who played in which position along the feeder channel. In phase 2, players knew who was in their group and in which position, and their individual decisions were publicly disclosed after every round. In addition, players were allowed to communicate and discuss in each round for up to five minutes before making their individual choices. All these changes were announced before phase 2 started, and groups and player positions were reshuffled to maintain the anonymity of phase 1.

Before the game began with phase 1, two practice rounds were played to ensure all players understood the game. At the start of each round, participants received an initial endowment of 3,000 play rupees in an orange envelope. They also received an empty red envelope. All participants decided simultaneously on the amount each of them wanted to invest in dam maintenance, which they placed in their red envelope. All money not invested remained in the orange envelopes as private savings accounts for each

player. In the next step, both envelopes were returned to the field assistants, who announced and depicted the total group investment. Then participants made their crop choice, checking either the unsustainable or sustainable crop on a decision card. These decision cards were collected by the field assistants in another envelope to ensure anonymity. Finally, the field assistants checked whether each player could grow their chosen crop, and earnings for each player were calculated based on their private savings from decision 1 and the returns from their crop choice if the player was able to grow the crop. In phase 1, players receive a personal sheet containing information about their investment and round earnings depending on their chosen crop. This maintained anonymity but allowed players to keep track of the consequences of their investment and crop decisions. In phase 2, contributions, crop choices, and earnings of all players were written on a poster visible to all participants.

After all 10 rounds were played, the game ended with a debriefing. All participants and other community members could discuss the experience of the game for approximately 15 minutes. The facilitators asked open questions intended to relate the game experience to the community's real-life water challenges.

To test whether incentivized payments matter, half the sites were randomly selected to receive individual payments for the players based on their winnings in the game; the other half received a lump sum contribution to the community, but players were not given individual payments.

The complete protocol of the game is open accessible at:
https://gamesforsustainability.org/practitioners/#surface-water-game

The data on the rollout of the game in Rajashtan is open accessible under: https://doi.org/10.21421/D2/MFT8ZD

The data on the rollout of the game in Madhya Pradesh is open accessible under: https://doi.org/10.21421/D2/MFT8ZD

Table A1.2. Key features of the surface water game.

Design Features	**Explanation**
Game narrative and experience	
Theory of change	The citizens in the research areas increasingly rely on surface water structures for irrigation but face degraded infrastructure. This leads to water scarcity, especially in the post-rainy season. Collective maintenance investments and efficient water use through water-saving crops may increase community-level production and income.
Embedded or isolated intervention	The game was played at some sites where the NGO Foundation for Ecological Security (FES) was already implementing other water management interventions over the long term and in some sites where FES was not active.
Story framing	The story framing was very contextual. It was developed with the local NGO team to reflect the water management challenges of the communities in Rajasthan and Madhya Pradesh, India.
Goal definition	The goal was defined as generating individual income. It was explained that uncooperative choices decrease the group income and lead to unequal

	benefits.
Facilitation	Local community members were trained in facilitating the games and debriefings. They were requested to interfere as little as possible.
Game flow	In the facilitation training, emphasis was placed on good coordination among the facilitation team, which considerably reduced the time that players sat idle.
Graphics of posters and game materials	Illustrations including drawings, reflecting the regional context, were used to explain the game. Understanding the material did not require the ability to read.
Language	During pre-tests, the explanations were repeatedly revised with the local field team to ensure they are well understood.
Integration of any embodiment	No expression of ideas, quality, or feelings in a tangible or visible form was included int the intervention.
Game length	Each game session with debriefing took between 1.5 and 2 hours.
Debriefing	After each game session, there was a 15-minute debriefing allowing players and other community members to discuss how the game related to their real-life water management challenges.
Group competition	Two groups played simultaneously and could observe each other. The games of the two groups where not linked, but the situation often created an informal competition.
Game rules	
Response options	In line with our theory of change, the players could make two decisions: a) invest a share of their endowment in infrastructure maintenance and b) choose between water-intensive and water-saving crops. In Rajasthan, after the first 10 game rounds, players were given the opportunity to play a game variation based on the issues brought up during discussion between the rounds. Such variations included a) the possibility to sanction players, b) response to uncertain external support, c) the effect of path dependencies, and d) unequal endowments and benefits.
Payoff structure	We conducted a pre-assessment of real-life infrastructure maintenance costs and of benefits received from irrigation together with local FES staff. The game payoffs were accordingly adjusted to reflect the costs and benefits typical for the research area.
Uncertain payoffs	Payoffs were fixed without a random function.
Sequence of decisions	Decisions on the contributions to infrastructure maintenance were made simultaneously. Crop choices were made sequentially with subsequent players only being able to use water if the upstream players left sufficient water.
Repetition of decision	Ten rounds were played per session. There was no path dependency and the rounds where independent from each other.
Communication	Communication was not allowed during the first 5 rounds. In rounds 6 to 10, discussion was encouraged.
Consequences	Including punishment in the game was an optional game variation in the Rajasthan version of the game.
Encouragement	In rounds without communication, players received open feedback on the group performance and hidden feedback on individual earnings. In rounds with communication, all players' decisions as well as individual and group

	earnings were posted for everybody to see.
Power	Our game did not specifically address power issues, although the sequential water access represents a power dynamic. The optional game variation with unequal endowments and benefits also touches on common power imbalances.
Player attributes	
Personality and identity	The groups included both men and women together, with a varying share of women.
History/experience with management challenge	Community members who live downstream from a community dam were the target group of the game. In real life, this group experiences degraded common water infrastructure. If water is available for irrigation, their crop choices affect water availability further downstream.
Group size	The Rajasthan game was played with 10 players. The Madhya Pradesh version of the game was played with 2 groups of 7 players each.

Forestry Game

This game was designed to simulate the social dilemmas around community forest management that arise from the tension between private earnings from harvesting and collective resource sustainability. This problem context is translated into the game context by presenting players with one main decision: how many trees to harvest in their community forest. If players harvest the trees too fast and stocks fall below a certain threshold, the forest loses its capacity to regenerate and will be depleted, leaving players without future revenues from the forest. However, players have an incentive to harvest as many trees as possible as each tree provides revenue of INR 5. Choosing a sustainable number of trees to harvest yields lower returns per time period but a stable time horizon securing long-term benefits for users. Thus, the socially optimal strategy is for all players to choose a sustainable harvest level that is compatible with its regeneration rate, allowing the forest to maintain (or even expand) its stock. This maximizes earnings at the group level over the long run.

The game starts with a total of 50 trees available in the shared forest. In the successive rounds of play, each player chooses (privately) how many trees to harvest. There is a maximum number of trees that a player can harvest in each round depending on resource size at the beginning of each round. Each harvested tree yields INR 5. The facilitators record the individually chosen number of harvested trees and pay the respective earnings in play money to each player. The aggregate harvest of all players is then calculated. Finally, the stock is regenerated by adding 1 tree for every 10 trees remaining. The new forest stock is announced to the group and the next round starts, unless the maximum number of rounds have been reached. If the remaining tree stock level falls below 5 units, there are not enough trees for the group, and game ends.

The game was played in 60 randomly chosen sites in Andhra Pradesh and Rajasthan. At each site, two groups played in parallel, one women's group and one men's group. Each group consisted of five players who played the game in a relatively "closed" setting, meaning typically there were no other observers from the community. Each group played a maximum of 21 rounds. The rounds were divided into three phases. Within each phase, there was a fixed maximum number of rounds, although this number was not announced. However, if the forest was depleted before reaching the maximum number of rounds, the phase would end, and the next phase started.

Before the game began with phase 1, three practice rounds were played to ensure all players understood the game. In phase 1, participants played without communication. After a maximum of six rounds, or when the forest was depleted, game phase 1 was over and players were allowed to discuss what happened during the rounds and what they would want to do to improve the outcome. In phase 2, they played a second set of up to six rounds, with discussion among the players allowed before making their individual choices. Again, after a maximum of seven rounds or when the forest was depleted, game phase 2 was over. In phase 3, they played a final set of up to seven rounds or until the forest was depleted, with communication allowed and optional rule election. For the rule election, players were asked whether they want to introduce monitoring rules based on their experience in the first two phases. If the group decided not to introduce monitoring, they continued as in the second phase with communication time between each round. If the group decided to introduce monitoring, they spent up to 10 minutes discussing what kind of monitoring mechanisms they would like to have, and notes were taken of the suggested ideas as well as their final chosen monitoring rule. After three rounds played with the monitoring mechanism, the groups were asked whether they wanted to introduce some form

of penalty for the violators (i.e., sanctioning). If the groups declined, they continued for a further four rounds with monitoring. If they agreed, the groups discussed what kind of punishment they would like to introduce and then played a further four rounds incorporating their chosen sanctioning method.

Typically, on the next day, a debriefing was held for the whole community, at which a facilitator described what had been happening with regards to the local community forest, players discussed their experiences in the game, and the community discussed implications for their community.

To test whether incentivized payments matter, half the sites were randomly selected to receive individual payments for the players based on their winnings in the game; the other half received a flat lump sum contribution to the community, but players were not given individual payments.

To test the effect of visual reminders of ecosystem services from forests, half the sites were randomly selected to use a white display board that showed the spatial layout of the resources (trees) in each round; the other half used a color display board with different colors linked to different resource sizes (i.e., forest conditions) with varying levels of ecosystem services provided by the forest.

The complete protocol of the game is open accessible at:
https://gamesforsustainability.org/practitioners/#forest-game

The data of rollout of the game in Andhra Pradesh and Rajasthan in 2017 and 2018 is accessible under:
https://dataverse.harvard.edu/privateurl.xhtml?token=3a2861be-3f71-4601-9e78-4fb235982387

Table A1.3. Key features of the forestry game.

Design Features	**Explanation**
Game narrative and experience	
Theory of change	The citizens in the research areas experience forest degradation, which can be addressed by improved collective action and common-pool resource (CPR) management institutions.
Embedded or isolated intervention	Games were played at sites in Andhra Pradesh and Rajasthan where the NGO Foundation for Ecological Security (FES) was already engaging with the communities through its commons program.
Story framing	The story framing was very contextual. It was developed with the local NGO team to reflect the forest management challenges of the communities in this area and highlight the tensions between private earnings and collective resource sustainability.
Goal definition	The goal was defined as generating individual income. The game parameters were calibrated such that uncooperative choices deplete the forest and decrease the group income.
Facilitation	FES members were trained in facilitating the games and debriefings. They were requested to interfere as little as possible.
Game flow	Facilitation training emphasized good coordination among the facilitation team, which reduced the time players sat idle.
Graphics of posters and game materials	Two different game boards were used: a simple black and white one and a coloured one indicating ecosystem services from denser forests. Understanding the material did not require the ability to read.

Language	During pre-tests, the explanations were repeatedly revised with the local field team to ensure that they are well understood. The field teams explained the game in a local dialect.
Integration of any embodiment	No expression of ideas, quality, or feelings in a tangible or visible form was included int the intervention.
Game length	Each game session took approximately 1.5 hours. The debriefing on the next day typically took 1.5 hours.
Debriefing	Typically, on the next day after playing the game, all members of the site were invited to participate in a community debriefing that allowed players and other community members to discuss how the game related to their real-life CPR management challenges.
Group competition	The games were held at separate places for men and women at the same time. The two groups could not observe each other.
Game rules	
Response options	In line with our theory of change, the players made one decision—the number of trees to harvest—which directly impacted the future stock of the forest and income. In game phase 2, players were additionally allowed to communicate. Phase 3 allowed players to elect optional rules for CPR management, including the possibility of sanctions.
Payoff structure	The game payoffs were calibrated such that the average earnings of the players under the individual payment method were equal to about a half day of wage labor.
Uncertain payoffs	Payoffs depended on the number of trees harvested under the individual payment method.
Sequence of decisions	Decisions on the tree harvest were made simultaneously.
Repetition of decision	A maximum of 21 rounds were played per session, divided into three phases. Phases were independent from each other. However, rounds within a phase had a path dependency regarding the harvested trees.
Communication	We asked players not to talk during the first game phase. In game phases 2 and 3, discussion was allowed.
Consequences	Including punishment in the game was up to the groups to decide in game phase 3.
Encouragement	In all rounds, only group performance was openly revealed on a game board for everybody to see. In phase III, individual choices were also revealed to allow the players to monitor rule compliance and enforced the rules.
Power	Our game did not specifically address power issues.
Player attributes	
Personality and identity	Groups were divided by gender.
History/experience with management challenge	Community forest users were the target group for the game. In real life, this group experiences degraded forest conditions.
Group size	The game was played at each site with two groups of five players each. One group included only men, the other group only women.